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October 31, 2022  
NRC-22-0043

10 CFR 50.73

U.S. Nuclear Regulatory Commission  
Attention: Document Control Desk  
Washington, DC 20555-0001

Fermi 2 Power Plant  
NRC Docket No. 50-341  
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2022-004

Pursuant to 10 CFR 50.73(a)(2)(i)(B), DTE Electric Company (DTE) is submitting LER No. 2022-004, Containment Isolation Valve Actuator Found Disconnected from its Bonnet Resulting in Condition Prohibited by Technical Specifications.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

A handwritten signature in black ink, appearing to read "PD", written over a faint circular stamp.

Peter Dietrich  
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2022-004, Containment Isolation Valve Actuator Found Disconnected from its Bonnet Resulting in Condition Prohibited by Technical Specifications.

cc: NRC Project Manager  
NRC Resident Office  
Regional Administrator, Region III

**Enclosure to  
NRC-22-0043**

**Fermi 2 NRC Docket No. 50-341  
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2022-004  
Containment Isolation Valve Actuator Found Disconnected from its  
Bonnet Resulting in Condition Prohibited by Technical Specifications**



## LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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## 1. Facility Name

Fermi 2

## 2. Docket Number

05000

341

## 3. Page

1 OF 4

## 4. Title

Containment Isolation Valve Actuator Found Disconnected from its Bonnet Resulting in Condition Prohibited by Technical Specifications

| 5. Event Date |     |      | 6. LER Number |                   |              | 7. Report Date |     |      | 8. Other Facilities Involved |               |
|---------------|-----|------|---------------|-------------------|--------------|----------------|-----|------|------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Revision No. | Month          | Day | Year | Facility Name                | Docket Number |
| 06            | 27  | 2022 | 2022          | - 004 -           | 00           | 10             | 31  | 2022 | N/A                          | 05000         |
|               |     |      |               |                   |              |                |     |      | Facility Name                | Docket Number |
|               |     |      |               |                   |              |                |     |      | N/A                          | 05000         |

## 9. Operating Mode

1

## 10. Power Level

9

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

| 10 CFR Part 20                                                           | <input type="checkbox"/> 20.2203(a)(2)(vi)  | <input type="checkbox"/> 50.36(c)(2)                  | <input type="checkbox"/> 50.73(a)(2)(iv)(A)   | <input type="checkbox"/> 50.73(a)(2)(x)  |
|--------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| <input type="checkbox"/> 20.2201(b)                                      | <input type="checkbox"/> 20.2203(a)(3)(i)   | <input type="checkbox"/> 50.46(a)(3)(ii)              | <input type="checkbox"/> 50.73(a)(2)(v)(A)    | 10 CFR Part 73                           |
| <input type="checkbox"/> 20.2201(d)                                      | <input type="checkbox"/> 20.2203(a)(3)(ii)  | <input type="checkbox"/> 50.69(g)                     | <input type="checkbox"/> 50.73(a)(2)(v)(B)    | <input type="checkbox"/> 73.71(a)(4)     |
| <input type="checkbox"/> 20.2203(a)(1)                                   | <input type="checkbox"/> 20.2203(a)(4)      | <input type="checkbox"/> 50.73(a)(2)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(v)(C)    | <input type="checkbox"/> 73.71(a)(5)     |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                | 10 CFR Part 21                              | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) | <input type="checkbox"/> 50.73(a)(2)(v)(D)    | <input type="checkbox"/> 73.77(a)(1)(i)  |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                               | <input type="checkbox"/> 21.2(c)            | <input type="checkbox"/> 50.73(a)(2)(i)(C)            | <input type="checkbox"/> 50.73(a)(2)(vii)     | <input type="checkbox"/> 73.77(a)(2)(i)  |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                              | 10 CFR Part 50                              | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(viii)(A) | <input type="checkbox"/> 73.77(a)(2)(ii) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                               | <input type="checkbox"/> 50.36(c)(1)(i)(A)  | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |                                          |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                | <input type="checkbox"/> 50.36(c)(1)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(iii)             | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |                                          |
| <input type="checkbox"/> OTHER (Specify here, in abstract, or NRC 366A). |                                             |                                                       |                                               |                                          |

## 12. Licensee Contact for this LER

## Licensee Contact

Eric Frank – Manager, Nuclear Licensing

## Phone Number (Include area code)

734-586-4772

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| A     | JM     | ISV       | L200         | YES                |       |        |           |              |                    |

## 14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

## 15. Expected Submission Date

Month Day Year

## 16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 06/27/2022 during start up walkdowns for Forced Outage 22-01, engineering personnel discovered the B2103F019 (Main Steam Outboard Drain Isolation valve) motor operator separated from the valve bonnet yoke arm flange. The motor operator was energized (motor and gear train freely running) and agitating on the end of the vertically oriented valve stem. The inboard isolation valve (B2103F016) was capable to isolate during the entire time the B2103F019 was inoperable. Upon identification, the Containment Isolation Valve was declared inoperable per Technical Specification 3.6.1.3. The cause was determined to be improper thread engagement, due to personnel error of installing the incorrect bolts following testing using an installed testing device. B2103F019 was secured and is planned to be reworked during the next availability (refueling outage). The cycle schedule was reviewed for the impact of B2103F016 and B2103F019 isolated with no issues identified. An extent of condition was performed on the other motor operated valves tested during the previous refueling outage.

This event is being reported under 10 CFR 50.73(a)(2)(i)(B) for operation or condition prohibited by TS.



**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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**1. FACILITY NAME**

Fermi 2

**2. DOCKET NUMBER**

05000- 341

**3. LER NUMBER**

YEAR

2022

SEQUENTIAL  
NUMBER

004

REV  
NO.

00

**NARRATIVE****INITIAL PLANT CONDITIONS**

Mode – 1

Reactor Power – 9 Percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

**DESCRIPTION OF THE EVENT**

On 06/27/2022 during start up walkdowns for Forced Outage 22-01, engineering personnel discovered the B2103F019 (Main Steam Outboard Drain Isolation valve) [SB][ISV] motor operator [84] separated from the valve bonnet yoke arm flange [V]. The motor operator was energized (motor and gear train freely running) and agitating on the end of the vertically oriented valve stem. Having run out of stem threads, the actuator was being restrained on the end of the valve stem by the unthreaded portion of the stem nut counterbore and by the electrical conduit that serves the motor operated valve (MOV). The MOV was de-energized by shutting off the actuator power supply at the Motor Control Center.

During the initial investigation, the (4) actuator mounting bolts had pulled out of the actuator housing, with visible damage to the threaded actuator mounting holes. After inspection it was identified that the removed bolts were 1 in long 5/16 inch cap bolts that, when fully tightened, only had 2 or 3 threads of engagement into the actuator.

Based on the sequence of events, as noted per plant computer data, during the 06/27/2022 closing stroke the actuator-to-bonnet yoke flange-threaded connections failed. This allowed the unrestrained actuator to traverse up the valve stem as the stem nut rotated about the valve-stem threads.

**SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS**

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

B2103F019 is a normally closed Primary Containment Isolation Valve (PCIV) [JM] and has a safety function to automatically close on a Reactor Vessel Low Level 1, and on Accident Isolation Signals E- Main Steam Line High Flow, F- Main Steam Line Tunnel High Temperature, G- Main Steam Line Low Pressure, J- Low Condenser Vacuum, P- Turbine Building High Temperatures to provide primary containment isolation of penetration, X-8. B2103F019 does not have a safety function in the open position. It is opened to provide a drain path for Main Steam line moisture to the turbine drain system during plant start-up or similar conditions. It is an IST Appendix III (formerly NRC GL 96-05) MOV with an active safety function to close within a design basis stroke time of 23 seconds per TRM 3.6.3 Table TR3.6.3-1 Group 1 Main Steam Automatic Isolation Valves. It is Type C Leak Tested and Bypass Leakage Tested under the Appendix J program. It is classified as Seismic Class I.



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| 1. FACILITY NAME | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|------------------|------------------|---------------|-------------------|---------|
| Fermi 2          | 05000- 341       | YEAR          | SEQUENTIAL NUMBER | REV NO. |
|                  |                  | 2022          | 004               | 00      |

**NARRATIVE**

Based on physical condition of valve B2103F019, it would not have been able to close and fulfill its function to isolate primary containment. For penetration X-8, the inboard containment isolation valve (B2103F016) was Operable throughout the entire period, where B2103F019 was not. While redundancy of the primary containment isolation function was compromised, penetration X-8 was able to be isolated.

Although the penetration isolation of the Main Steam drain line safety function was maintained operable (based on the B2103F016 being capable and operable to isolate during the entire time B2103F019 was inoperable), TS 3.6.1.3 states that each PCIV (except vacuum breakers) shall be operable in Modes 1, 2, and 3. Condition A identifies that if the penetration flow path with one inoperable PCIV, then the flow path shall be isolated by closing and deactivating at least one automatic or manual valve in the flow path within 4 hours. The initial past operability for the initial discovery of the issue on 6/27/2022 determined the condition was not reportable based on the Safety Function of the flow path was maintained by B2103F016 being operable. The past operability failed to identify that this condition is prohibited by Technical Specifications and B2103F019 was not operable as an isolation valve since the startup of the Refueling outage.

Probabilistic Risk Analysis was performed on the inoperable B2103F019, and it was determined that there was no change in Core Damage Frequency (CDF), and minimal change in Large Early Release Frequency (LERF).

**CAUSE OF THE EVENT**

The cause was determined to be improper thread engagement, due to personnel error (utility employees) of installing the incorrect bolts following testing using an installed testing device. Thread engagement was not achievable due to bolt length. The WO review concluded that bolting was ordered to perform thrust test utilizing a Torque-Thrust Cell (TTC). Per Crane Nuclear documentation, (4) 5/16-18 x 1-inch cap screws are required to mount a RA TTC to an SMB-000 actuator and the final installed bolting for the yolk/actuator is 5/16-18 x 1-1/2-inch. Maintenance activities during the refueling outage used shorter testing bolts during final assembly to secure the actuator to the valve yoke flange which resulted in improper thread engagement (as found actuator bolting was confirmed to be 5/16- 18 x 1-inch cap screws) .

**CORRECTIVE ACTIONS**

B2103F019 was secured and is planned to be reworked during the next availability (forced or refueling outage). The cycle schedule was reviewed for the impact of B2103F016 and B2103F019 isolated for the remainder of the cycle with no issues identified.

Seven additional Motor Operated Valves were identified as being in a potential similar condition due to the mounting bolts being removed during the previous refueling outage for maintenance. On six of the seven, a visual inspection determined that the correct bolting is present, and no separation exists between the actuator and bonnet as described in the condition description. The seventh valve (B2103F016) is located in the drywell and is inaccessible. As described above, B2103F016 is the inboard isolation valve in series with B2103F019, B2103F016 has operated a minimum of six times since the previous refueling outage and has not shown any indication of a similar issue. B2103F016 has been deenergized in the closed position to isolate the containment penetration X-8.



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|                  |  |                  |  | 2022          | 004               | 00      |

**NARRATIVE****PREVIOUS OCCURRENCES**

A review of the Corrective Action Program for additional Containment Isolation Valves that may have been inappropriately classified as not reportable as the safety function was not challenged due to the subsequent in series valve was operable was performed. Three issues were correctly identified at time of discovery during the surveillance requirement and repaired. Another issue identified by a similar surveillance requirement (but was monitored at an increased surveillance frequency) was previously identified by the NRC under Green NCV 05000341/2022001-02. This issue was an NRC violation as the subsequent increased frequency was not performed as stated.